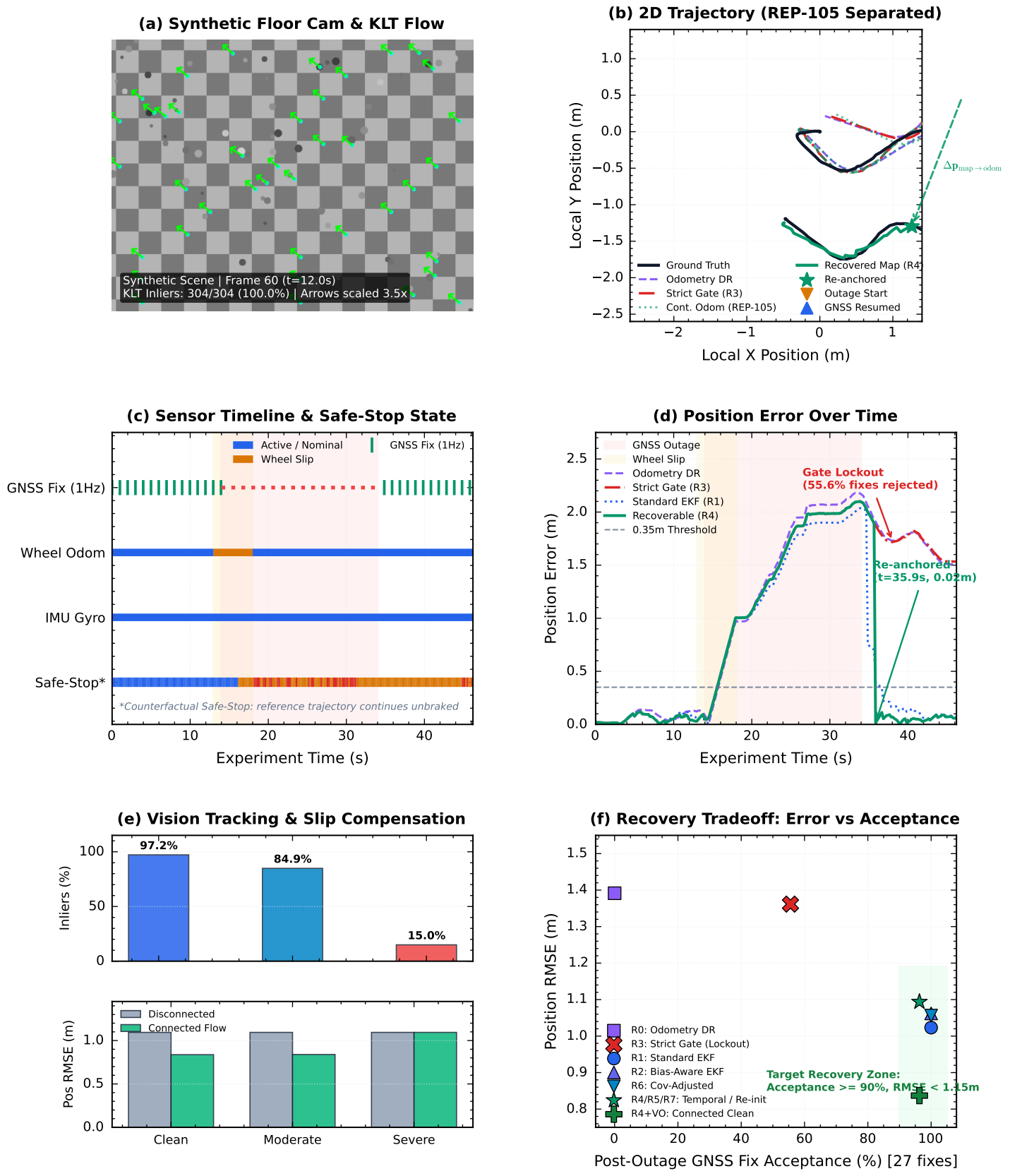

(a) Synthetic Floor Cam & KLT Flow
Synthetic Scene | Frame 60 (t=12.0s)
KLT Inliers: 304/304 (100.0%) | Arrows scaled 3.5x
(b) 2D Trajectory (REP-105 Separated)
Local Y Position (m)
Local X Position (m)
Ground Truth
Odometry DR
Strict Gate (R3)
Cont. Odom (REP-105)
Recovered Map (R4)
Re-anchored
Outage Start
GNSS Resumed
$\Delta p_{map \to odom}$
(c) Sensor Timeline & Safe-Stop State
Active / Nominal
Wheel Slip
GNSS Fix (1Hz)
GNSS Fix (1Hz)
Wheel Odom
IMU Gyro
Safe-Stop*
*Counterfactual Safe-Stop: reference trajectory continues unbraked
Experiment Time (s)
(d) Position Error Over Time
GNSS Outage
Wheel Slip
Odometry DR
Strict Gate (R3)
Standard EKF (R1)
Recoverable (R4)
0.35m Threshold
Gate Lockout
(55.6% fixes rejected)
Re-anchored
(t=35.9s, 0.02m)
Position Error (m)
Experiment Time (s)
(e) Vision Tracking & Slip Compensation
Inliers (%)
97.2%
84.9%
15.0%
Pos RMSE (m)
Disconnected
Connected Flow
Clean
Moderate
Severe
(f) Recovery Tradeoff: Error vs Acceptance
Position RMSE (m)
R0: Odometry DR
R3: Strict Gate (Lockout)
R1: Standard EKF
R2: Bias-Aware EKF
R6: Cov-Adjusted
R4/R5/R7: Temporal / Re-init
R4+VO: Connected Clean
Target Recovery Zone:
Acceptance >= 90%, RMSE < 1.15m
Post-Outage GNSS Fix Acceptance (%) [27 fixes]